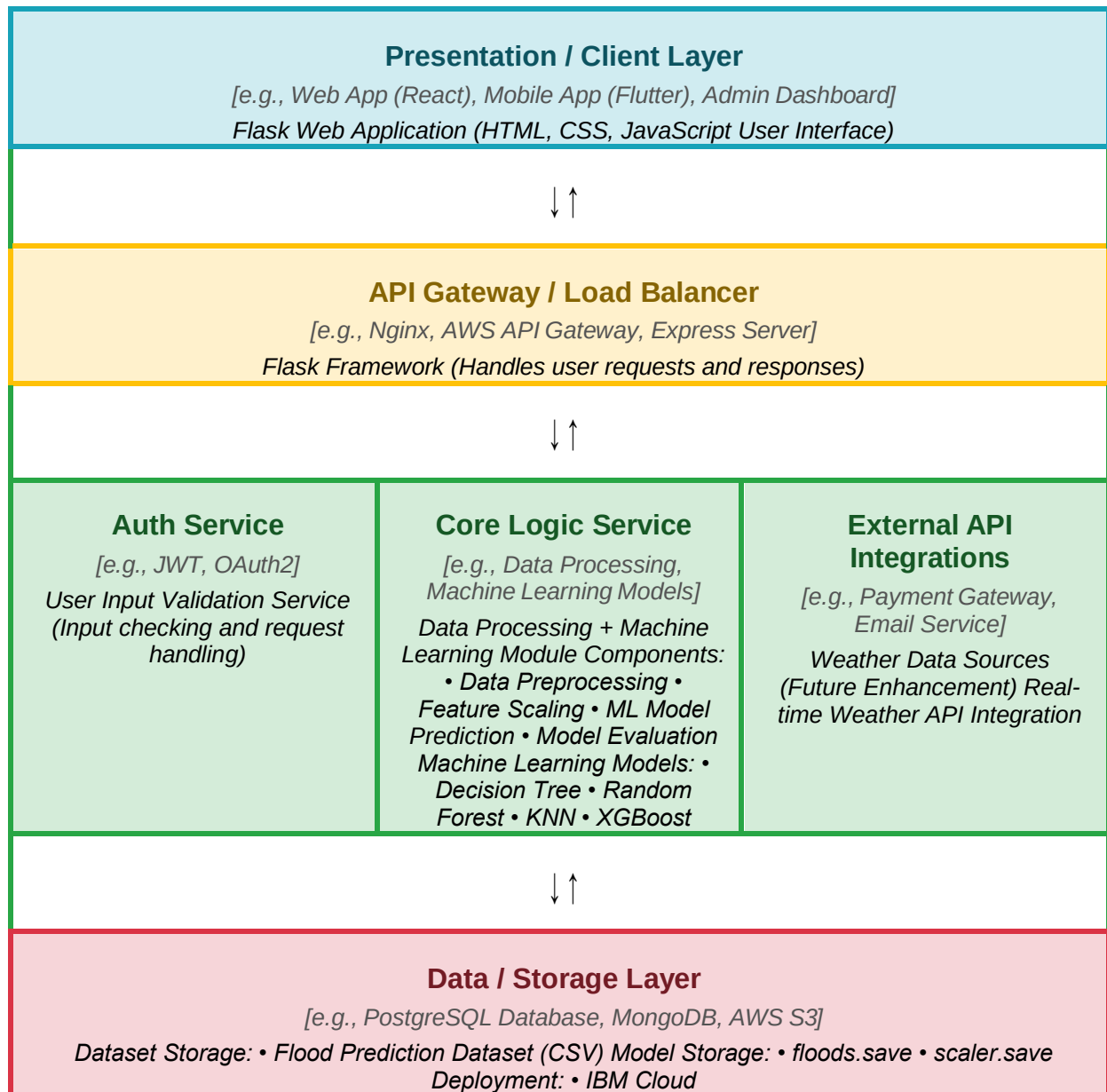


Rising Waters: A Machine Learning Approach to Flood Prediction

Solution Architecture

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where users enter weather parameters and view flood prediction results.	HTML, CSS, JavaScript, Flask Templates
API Gateway / Application Server	Handles communication between the user interface and machine learning backend. Receives requests and returns prediction responses.	Flask Framework, Python
Authentication Service	Validates user inputs and ensures correct data format before prediction processing.	Python Validation Logic
Core Logic Service	Performs data preprocessing, feature transformation, loads ML models, and generates flood predictions.	Python, NumPy, Pandas, Scikit-learn, XGBoost